

AE4889
Special Topics in Astrodynamics
MIMOSA – MIssion to Mars with a sOlar SAil

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github.com/dptrias/SpecialTopicsAstrodynamics

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Personal problem specification

Parameter	Value
Sun–Earth Lagrange point	L_2
Sun–Mars Lagrange point	L_2
Shift in Lagrange points	0.0005
Radius Martian parking orbit	306 666 km
Lightness number	0.025

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List of Acronyms

AEP Artificial Equilibrium Point. [2](#)

CR3BP Circular Restricted Three-Body Problem. [1](#)

JPL Jet Propulsion Laboratory. [1](#)

WP1 Sun–Earth and Sun–Mars Lagrange points

1.1a The Lagrange points of the Sun–Earth and Sun–Mars systems are computed using Newton’s method over equation (1), which is derived from the [Circular Restricted Three-Body Problem \(CR3BP\)](#). The resulting dimensionless coordinates of the colinear Lagrange points (L_2) are presented in Table 1.

$$x - \frac{1 - \mu}{(x + \mu)^2} - \frac{\mu}{(x - 1 + \mu)^2} = 0 \quad (1)$$

Table 1: Dimensionless coordinates (x_i, y_i, z_i) of the Sun–Earth and Sun–Mars colinear Lagrange points in their respective synodic reference frames, obtained from the solution of the circular restricted three-body problem using a Newton method.

Lagrange point	x_i [LU]	y_i [LU]	z_i [LU]
Sun–Earth L_2	1.01009044	0	0
Sun–Mars L_2	1.00476311	0	0

1.1b For verification purposes, the dimensionless coordinates of the colinear Lagrange points are also extracted from the [Jet Propulsion Laboratory \(JPL\) Solar System Dynamics three-body periodic orbits tool](#), which are presented in Table 2. It can be observed that for the represented number of digits, the values are identical, with differences on the order of 10^{-9} LU, likely due to shared system parameters between both methods. *(53 words)*

Table 2: Dimensionless x coordinates of the Sun–Earth and Sun–Mars colinear Lagrange points in their respective synodic reference frames per JPL and their absolute differences w.r.t. the values obtained from the Newton method.

Lagrange point	x_i [LU]	Δx_i [LU]
Sun–Earth L_2	1.01009044	$4.21585256 \times 10^{-9}$
Sun–Mars L_2	1.00476311	$3.32491457 \times 10^{-9}$

1.2 From the results it can be observed that the Sun–Mars L_2 point is closer to Mars than the Earth L_2 is to Earth (in dimensionless units). At L_2 , the gravitational pulls of both bodies and the centrifugal force balance. Mars, being less massive, exerts a weaker gravitational pull, so the equilibrium point must sit closer to the planet to compensate, allowing its gravity to contribute sufficiently to the force balance. *(70 words)*

WP2 Artificial Lagrange points

2.1a The required lightness number, β , to achieve a shift of 0.0005 LU in the Lagrange points along the Sun-Planet line is computed using the formula derived from the balance of forces at the artificial equilibrium point (lying at the Sun-Planet line):

$$\beta = \frac{(x_{\text{AEP}} + \mu)^2}{1 - \mu} \left[-x_{\text{AEP}} + \frac{1 - \mu}{(x_{\text{AEP}} + \mu)^2} + \frac{\mu}{(x_{\text{AEP}} - (1 - \mu))^2} \right] \quad (2)$$

which produces the results shown in Table 3.

Table 3: Required lightness number to achieve the specified shift in the Lagrange points.

Lagrange point	Shift in Lagrange point along the Sun-Planet line	Required lightness number, β
Sun–Earth L_2	0.0005	$4.76757628 \times 10^{-3}$
Sun–Mars L_2	0.0005	$5.06104189 \times 10^{-3}$

With respect to the feasibility of achieving such lightness numbers, previously proposed missions such as *Sunjammer* have targeted lightness numbers on the order of 0.03, [1], which is an order of magnitude higher than the required values, indicating that the proposed shifts are indeed feasible with current or near-term solar sail technology. (51 words)

2.1b Since the required shift is along the Sun-Planet line, and all the forces (gravitational and centrifugal) at the colinear Lagrange points are aligned along this line, the required sail orientation to achieve the desired shift is also aligned with this line, that is, with the sail normal vector pointing directly away from the Sun, as shown in Table 4. (58 words)

Table 4: Required sail orientation to achieve the required shift in the Lagrange points.

Lagrange point	$n_{x,i}$	$n_{y,i}$	$n_{z,i}$
Sun–Earth L_2	1	0	0
Sun–Mars L_2	1	0	0

2.2 To verify the required lightness numbers calculations the colinear [Artificial Equilibrium Point \(AEP\)](#) specified in [1] is calculated using this work’s implementation and compared to the value reported in the reference, To verify the required lightness numbers calculations the colinear [Artificial Equilibrium Point \(AEP\)](#) specified in [1] is calculated using this work’s implementation and compared to the value reported in the reference, the considered case being the Sun–Earth system with a AEP located at $x_{\text{AEP}} = 0.983867$ LU. The resulting lightness number is $\beta = 0.0363$, which matches the value reported in [1], confirming the correctness of the implementation. (64 words)

2.3 The required area of the solar sail is derived from the definition of the lightness number:

$$\beta \frac{\mu_{\odot}}{r^2} = 2P \frac{A}{m} \rightarrow A = \frac{\beta \mu_{\odot} m}{2Pr^2} \quad (3)$$

where P is the solar radiation pressure at the location of the sail, which can be calculated from the solar constant, $S_{\odot}$ [2], and the speed of light, c , as follows:

$$P = \frac{S_{\odot}}{c} \left(\frac{\text{AU}}{r} \right)^2 \quad (4)$$

Table 5: Required solar-sail side-length to achieve the required shift in the Lagrange points.

Lagrange point	Required solar-sail side-length [m]
Sun–Earth L_2	17.6459
Sun–Mars L_2	18.1809

The resulting sizes required to achieve the specified shifts in the Lagrange points are shown in Table 5. These sizes are within the range of previously proposed solar sails(e.g., *Sunjammer* had a sail area of $38 \times 38 \text{ m}^2$), indicating that the required shifts are indeed feasible with current or near-term solar sail technology.

2.4 To analyse the sensitivity of the required lightness number to the shift in the Lagrange point, a range of 21 shifts from 0.0005 LU to 0.001 LU is considered, and the corresponding lightness numbers are computed using the formula from Task 2.1a. The results are shown in Figure 1.

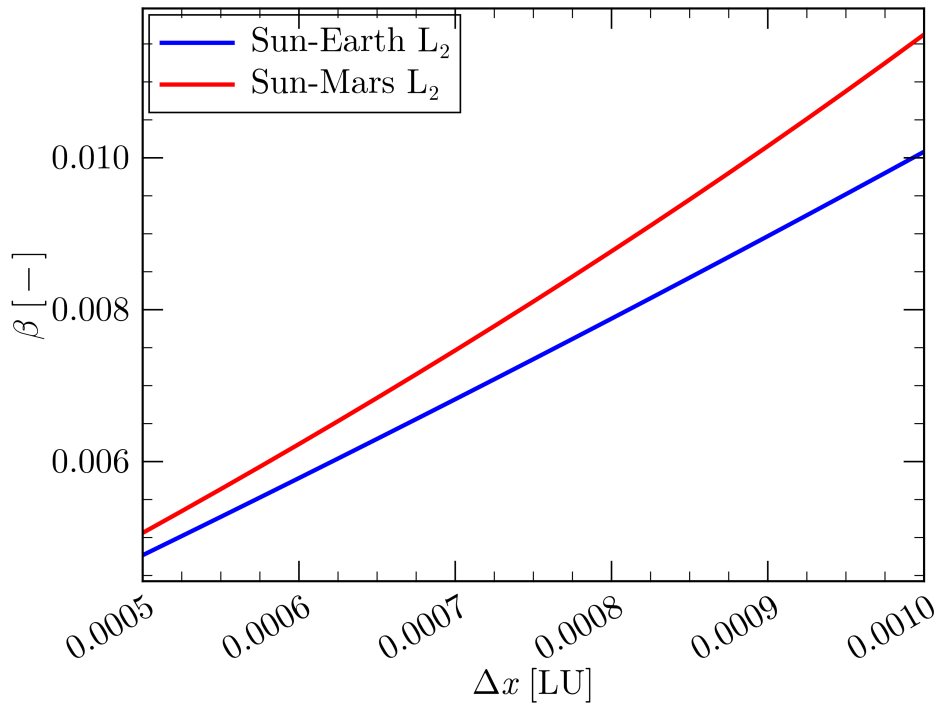


Figure 1: Sensitivity of the required lightness number, β , to the shift in the Lagrange point, Δx , for the Sun–Earth and Sun–Mars systems.

Here it can be observed that the required lightness number follows a quasi-linear trend with respect to the shift in the Lagrange point. As the shift Δx increases, the sail is displaced further from the natural L_2 location along the Sun-planet line, where the gravitational attractions of the Sun and the planet as well as the centrifugal acceleration all vary smoothly with position. For small displacements, these variations are well approximated as linear functions of Δx , so the additional solar radiation pressure acceleration required to maintain equilibrium at the new location also scales approximately linearly with the shift, producing the observed trend. (104 words)

WP3 Stability of the (artificial) Lagrange points

$$(\mathbf{a}_s)_{ij} = \frac{\partial a_{s,i}}{\partial j} = \zeta \hat{\mathbf{n}}^T \quad (5)$$

Where:

$$\zeta = \frac{4\beta(1-\mu)(\hat{\mathbf{n}} \cdot \hat{\mathbf{r}}_1)}{r_1^3} \left[\frac{\hat{\mathbf{n}}}{2} - (\hat{\mathbf{n}} \cdot \hat{\mathbf{r}}_1)\hat{\mathbf{r}}_1 \right] \quad (6)$$

Table 6: Eigenvalues of the classical and artificial Lagrange points.

Lagrange point	λ_1	λ_2	λ_3	λ_4
Sun–Earth Classical	−2.484281	−2.056992 <i>i</i>	+2.056992 <i>i</i>	+2.484281
Sun–Earth Artificial	−2.673872	−2.174388 <i>i</i>	+2.174388 <i>i</i>	+2.673872
Sun–Mars Classical	−2.496888	−2.064657 <i>i</i>	+2.064657 <i>i</i>	+2.496888
Sun–Mars Artificial	−2.931888	−2.335453 <i>i</i>	+2.335453 <i>i</i>	+2.931888

WP4 Invariant manifolds and application to Mars exploration mission

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References

- [1] J. Heiligers, B. Diedrich, W. Derbes, and C. McInnes, “Sunjammer: Preliminary end-to-end mission design,” Aug. 2014.
- [2] G. Kopp and J. L. Lean, “A new, lower value of total solar irradiance: Evidence and climate significance,” *Geophysical Research Letters*, vol. 38, no. 1,